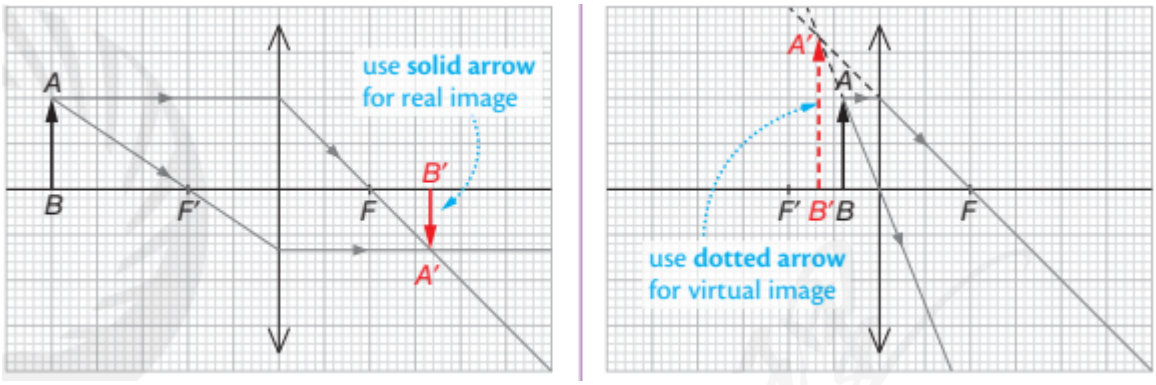
3.3.1 Real and virtual images

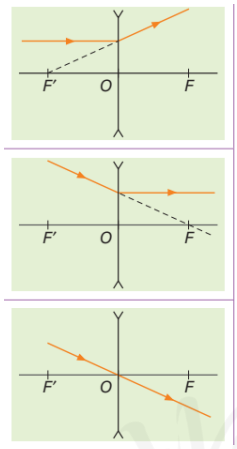
A real image is formed by actual convergence of rays and can be captured on a screen. A virtual image is formed where backward ray extensions meet. The rays do not pass through it, so it cannot be directly projected. Virtual images are still visible because light entering the eye appears to originate from their positions.



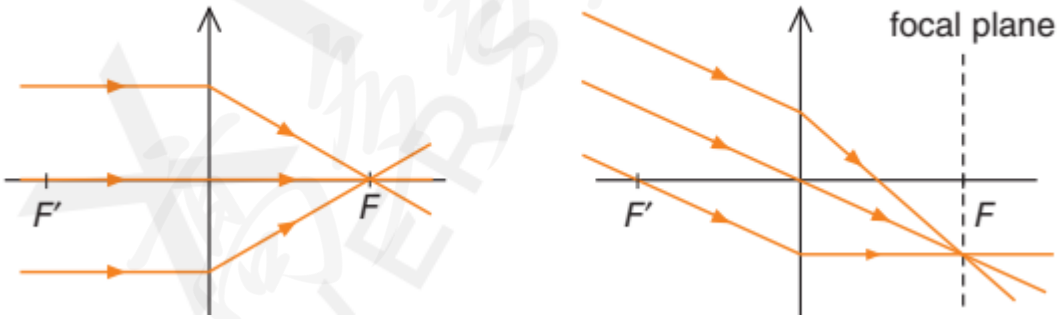
Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.

3.3.2 Principal rays of a concave lens

A concave lens diverges light. A ray parallel to the principal axis emerges as if from the near focus. A ray directed towards the far focus emerges parallel to the axis. A ray through the optical centre is approximately undeviated in the thin-lens model. Draw two rays from one object point and extend them backwards; their intersection locates the virtual image.



Three construction rays for a concave lens, showing the axis, focal points and ray directions.



Oblique parallel rays converge in the focal plane after passing through a convex lens.

3.3.3 Image formation by a convex lens

For a real object, a convex lens's image depends on object distance u and focal length f. When $u > f$, the image is real and inverted: diminished for $u > 2f$, equal-sized at $u = 2f$ and enlarged for $f < u < 2f$. For $u < f$, it is virtual, upright and enlarged on the object side. At $u = f$, emerging rays are parallel and no finite-distance image forms.

Draw any two of the three principal rays below from the same object point. Oblique parallel rays meet in the far focal plane: the plane through the focus perpendicular to the principal axis.

Light from one object point can pass through different parts of the lens to reach the same image point. Covering half the lens leaves the whole image at the same position, but makes it dimmer because less light passes through.

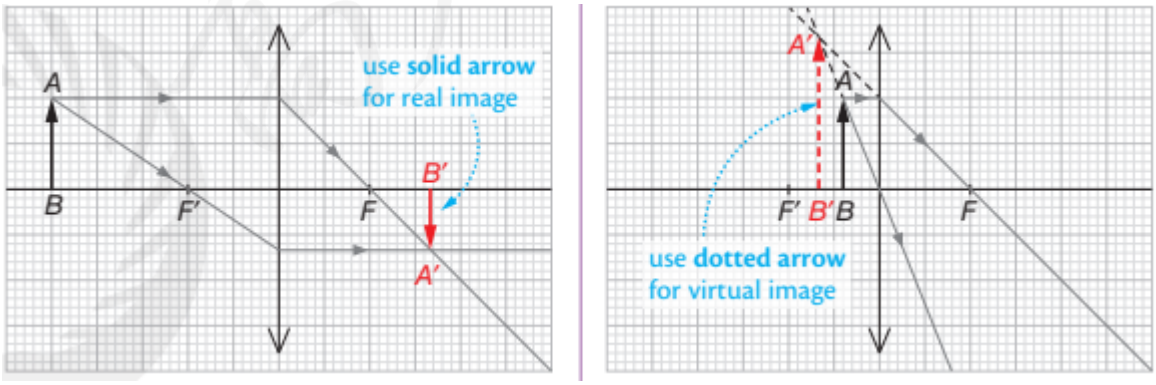
Incident ray	Emergent ray	Construction
Parallel to the principal axis	Through the far focus	Use the focus on the image side
Through the near focus	Parallel to the principal axis	Use the focus on the object side
Through the optical centre	Approximately undeviated	Thin-lens model

Linear magnification

Linear magnification compares image height with object height and can also be obtained from the magnitudes of image distance and real-object distance. Values above 1 mean enlargement and below 1 mean reduction. This card uses a non-negative size ratio, so orientation is stated separately.

$$m = \frac{h_i}{h_o} = \frac{|\,v\,|}{u}$$

Linear magnification as a size ratio is non-negative. h_i, h_o: image and object heights (same units); u > 0: real-object distance; v: signed image distance. The textbook's v/u uses distance magnitudes here; orientation is stated separately.



Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.

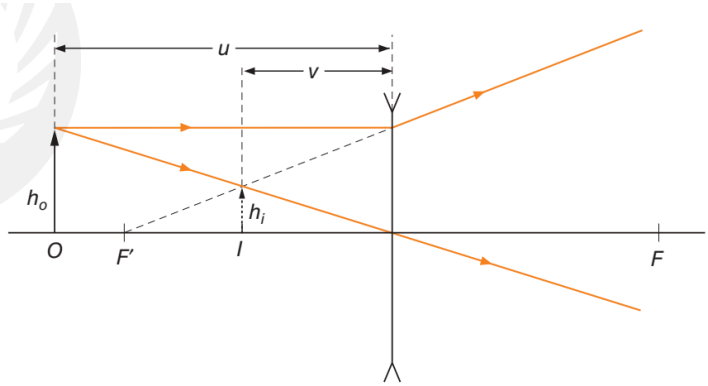


Fig 3.3e *Linear magnification of the image formed by a concave lens.*

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use $v < 0$.

3.3.4 Images formed by a concave lens

A concave lens diverges incident light. For a real object in front, backward extensions of emerging rays meet on the object's side, forming an upright, diminished virtual image between the lens and its near focal point. Use a ray parallel to the principal axis and a ray through the optical centre to locate it.

The thin-lens equation and signs

The thin-lens equation relates object distance, image distance and focal length for paraxial rays. This page takes real-object and real-image distances as positive, and virtual-image distance as negative. Focal length is positive for a convex lens and negative for a concave lens. Decide the image type and signs, then use consistent length units before substitution. For an object at infinity, reciprocal object distance is zero.

$$\frac{1}{u}+\frac{1}{v}=\frac{1}{f}$$

Thin lens, paraxial rays. Real object: u > 0; real image: v > 0; virtual image: v < 0. Converging lens: f > 0; diverging lens: f < 0. All distances in the same units.

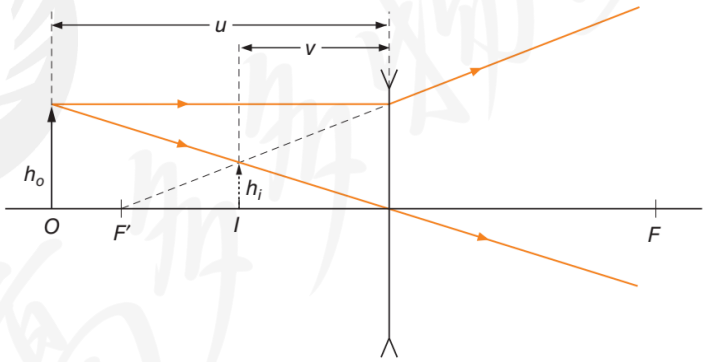
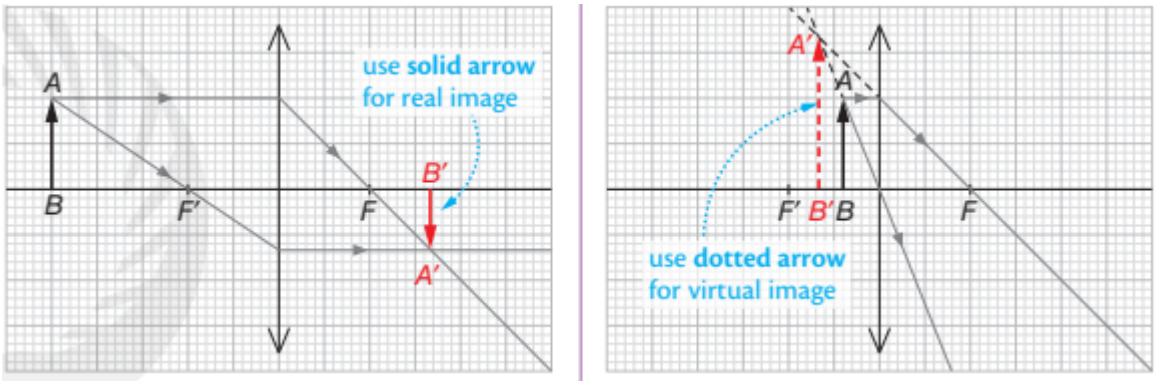


Fig 3.3e *Linear magnification of the image formed by a concave lens.*

Diverging rays trace back to a virtual, upright, smaller image on the object side. The diagram labels distance magnitudes; signed lens calculations use v < 0.



Left: an object beyond the focus forms an inverted real image where rays converge. Right: an object within the focal distance forms an upright virtual image where backward extensions meet.